

TerraBore™ MB-4400 Mining Blast Drill Rig

OEM Field Manual — Volume 1

AI-Augmented Field Engineer Edition

SECTION 1 — SAFETY & COMPLIANCE

1.1 General Safety

- Only certified field engineers may operate or service the MB-4400.
- Always perform Lockout/Tagout before accessing hydraulic or drill components.
- Maintain a minimum 5 m clearance around the drill mast during operation.
- Wear Class-IV mining PPE: respirator, vibration-rated gloves, reinforced boots.

1.2 Hazard Zones

- **Red Zone:** Active drilling radius (high-velocity debris).
- **Orange Zone:** Hydraulic line cluster (burst risk).
- **Yellow Zone:** Stabilizer outriggers (crush hazard).

1.3 AI-Assisted Safety

The AI Field Engineer Tool automatically:

- Flags unstable ground conditions using vibration mapping.
 - Predicts mast oscillation hazards.
 - Issues thermal overload warnings for hydraulic components.
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SECTION 2 — SYSTEM OVERVIEW

2.1 Machine Description

The TerraBore™ MB-4400 is a high-precision blast-drill rig designed for deep mining operations requiring accurate bore placement, high torque output, and stable drilling on uneven terrain.

2.2 Core Subsystems

- **Drill System:** BoreHead BH-2200 rotary assembly
 - **Hydraulic System:** HP-900 high-pressure pump
 - **Control System:** Ruggedized operator console
 - **Sensor Array:** Oscillation, depth, thermal, torque
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SECTION 3 — BILL OF MATERIALS (BOM)

3.1 Structural Assembly

- Chassis Frame (Reinforced Alloy-X)
- Stabilizer Outriggers (Hydraulic-Lock Type)
- Drill Mast Column (Tri-Segment Telescopic)

3.2 Drill System

- BoreHead BH-2200 Rotary Drill
- Carbide Drill Bit Set (Ø 150–300 mm)
- Torque Transfer Couplings (TT-Series)
- Vibration Dampening Rings (VD-44)

3.3 Hydraulic System

- HydroPump HP-900 (9000 PSI)
- Dual-Stage Pressure Regulator
- Hydraulic Line Set (High-Temp Rated)
- Fluid Reservoir (45 L, Heat-Shielded)

3.4 Sensor & Control

- Oscillation Sensor Pack (OS-7)
- Depth Mapping Array (DMA-4)
- Thermal Load Sensors (TLS-2)
- Control Console (Ruggedized)

3.5 Consumables

- Hydraulic Fluid HX-ThermaGrade
- Drill Bit Lubricant
- Dust Suppression Cartridges

SECTION 4 — OPERATING PROCEDURES

4.1 Startup Sequence

1. Engage stabilizer outriggers.
2. Power on the control console.
3. Run **Self-Calibration Cycle** (12 seconds).
4. Select drill bit size.
5. Initiate **Ground Stability Scan** (AI-assisted).
6. Begin drilling.

4.2 Drilling Workflow

- Maintain torque within 70–85% of rated load.
 - AI flags deviations above 0.8 mm oscillation.
 - Use dust suppression cartridges for deep bores.
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SECTION 5 — FIELD DIAGNOSTICS

5.1 Common Failures

- Mast oscillation
- Torque instability
- Hydraulic pressure drop
- Depth sensor drift

5.2 AI-Assisted Diagnostics

The AI tool provides:

- Oscillation heatmaps
- Torque waveform analysis
- Hydraulic leak prediction
- Depth sensor recalibration

5.3 Field Repair Procedure: Stabilizing Mast Oscillation

1. Activate **AI Oscillation Scan**.
 2. AI identifies oscillation source (mast segment, couplings, ground instability).
 3. Tighten mast segment clamps to AI-recommended torque values.
 4. Replace dampening rings if wear exceeds 40%.
 5. Re-run oscillation scan to confirm stability.
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SECTION 6 — MAINTENANCE

6.1 Daily Maintenance

- Inspect drill bit wear.
- Check hydraulic line temperature.
- Clean sensor array.

6.2 Weekly Maintenance

- Replace dust suppression cartridges.
- Flush hydraulic reservoir.
- Run AI-guided **Core Purge**.

6.3 Monthly Maintenance

- Replace vibration dampening rings.
- Perform full torque calibration.
- Update firmware via AI Fleet Console.

SECTION 7 — TROUBLESHOOTING MATRIX

Issue	Likely Cause	AI Diagnostic Output	Field Fix
Mast Oscillation	Loose clamps	Oscillation spike map	Tighten clamps, replace rings
Pressure Drop	Micro-leak	Acoustic leak signature	Replace hydraulic line
Depth Drift	Sensor contamination	Depth variance graph	Clean DMA-4 array
Torque Instability	Coupling wear	Torque waveform distortion	Replace TT-Series coupling

SECTION 8 — TECHNICAL SPECIFICATIONS

- **Torque Output:** 12,000 Nm
- **Max Bore Depth:** 180 m
- **Operating Temp:** -20°C to 65°C
- **Hydraulic Pressure:** 9000 PSI
- **Weight:** 14,200 kg
- **Power Source:** Diesel-Electric Hybrid